

## Supplementary Text

### The CSPdb

The CSPdb includes receptors that regulate diverse biological processes, including cell cycle progression (e.g., MAD2: 1KLQ), transcriptional control (e.g., Nrd1: 2MOW, 6GC3), translational regulation, autophagy, apoptosis (e.g., calcineurin: 2JOG; BCL-XL family: 2M04, 2LP8), and cell adhesion (e.g., tensin: 2LOZ). Disease-relevant targets address mechanisms in leukemia and other cancers (e.g., p53: 2MEJ), Alzheimer's disease (e.g., amyloid precursor protein-binding proteins: 2ROZ), and infectious diseases from viruses and bacteria, such as anthrax (2RUI), chikungunya (5I22), SARS-CoV-2 (7PKU), HIV (2L6E), and Ebola (2KQ0).

The CSPdb features several recurring protein domains that exhibit notable conformational changes upon ligand binding. For instance, it includes 23 interactions involving calmodulin (CaM) or CaM-like proteins with peptides (e.g., PDB IDs: 1SY9, 2JZI; full list in **SI Table S1**), which often involve substantial rearrangements upon binding described in detail elsewhere(1). Other receptors known for binding-induced conformational shifts or folding include the G protein Cdc42 (1CF4), the metaphase chromosome spindle checkpoint protein MAD2 (1KLQ), the KIX domain (2LXS), and CREB-binding proteins (e.g., 1R8U, 1L8C).

Additional recurring domains include the extra-terminal (ET) domains of bromodomain proteins BRD3 and BRD4 (e.g., 2NCZ, 2ND0; full list in **SI Table S7, SI Figure 6**), which bind intrinsically-disordered regions of viral integrase or chromatin regulator proteins. These interactions trigger subtle allosteric changes distal from the binding site, including the formation of a stabilizing secondary structure that reorients two allosteric helices (2, 3). Similarly, the PDZ2 domain (e.g., 1D5G, 1VJ6, 2M0V) shows allosteric effects mediated by the  $\beta$ 2– $\beta$ 3 loop during peptide recognition (4).

Nine entries involve the PH domain of the TFIIH transcription initiation factor subunit Tfb1 (e.g., 2LOX, 2M14; full list in **SI Table S8, SI Figure 7**). In these complexes, the ligand is often modeled as a helical polypeptide bound orthogonally to an extended  $\beta$ -sheet, resulting in propagation of allosteric signals via hydrogen-bond rearrangements. Other targets which involve hydrogen bond rearrangement in  $\beta$ -sheets include the Cdc42 (1CF4), Cyclophilin A (2MS4), and SH3 domain (e.g., 2OJ2, 2ROL; full list in **SI Table S1**). An additional eight entries use ubiquitin or ubiquitin-like (e.g. SUMO and NEDD8) protein receptors. Ubiquitin has been used as a model system for NMR studies (5) and for studying allostery (6) (e.g., 2KWV, 2MUR; full list in **SI Table S9, SI Figure 8**).

In Other Supplementary Information, you will find a All\_Case\_Studies.pdf, which provides a concise summary for each of the 139 targets in the CSPdb. Each case study contains the following panels: A) a  $^{15}\text{N}/^1\text{H}$  HSQC scatterplot overlaying the apo and holo assigned spectra

with the optimal N/H offset applied; B) Calculated CSPs for receptor residues along the sequence, bars colored by confusion matrix classification; C) the mask of significant CSPs; D) the mask of binding-site residues; E) the confusion matrix classification projected onto the structure: allosteric CSPs are purple (FP), CSPs in the binding site are green (TP), non-CSPs in the binding site are orange (FN), and non-CSPs outside of the binding site are blue (TN). Residues without CSP measurements are shown in gray; ligands are colored cyan.